Rohan Malhotra

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EDUCATION

New York University, Courant Institute of Mathematical Sciences — New York, NY
B.A. in Computer Science, Minor in Mathematics — *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications
- **Coursework:** Machine Learning, Linear Algebra, Probability & Statistics, Numerical Methods, Algorithms

EXPERIENCE

IBM — Dallas, TX
Applications Developer Intern, IBM Consulting — *May 2026 – Aug 2026*

- Fine-tuned Qwen-Coder-32B on Oracle EPM assets, raising EPM build-generation accuracy from **36.7% to 95.0%**.
- Built an on-prem EPM assistant that turns plain-language requests into reviewable JSON using RAG and MCP tools.
- Built an XGBoost cash-flow forecasting model and 13-day rolling cash forecasts in Oracle EPM Planning.

Robotics Engineer Intern — *Jun 2026 – Aug 2026*

- Trained YOLO11 models on 898 hand-labeled images to **99.5% mAP@50** with 99.4% precision and 99.4% recall.
- Ran real-time inference on Boston Dynamics Spot via OpenCV, gRPC, and lock-free queues; demoed at IBM DevCon.

Aress Software — New York, NY
Machine Learning Intern — *Jun 2025 – Aug 2025*

- Trained and deployed PyTorch models on **500K+ multimodal records** across Azure and AWS.
- Designed NoSQL pipelines unifying 10+ data sources into one analytics layer; optimized SQL queries for speed.

Kalshi — New York, NY
Quantitative Developer — *Jan 2026 – May 2026*

- Built Poisson hazard and Monte Carlo models of job-loss risk plus a Python/Next.js hedge-sizing engine.

Y-Axis Overseas Careers — Chatham, NJ
Data Engineering Intern — *May 2024 – Aug 2024*

- Engineered 12 predictive features from dataset analysis and transformations, raising model accuracy by **10%**.
- Cleaned and standardized 20K+ SQL records with schema fixes and validation, cutting data-prep time by **15%**.

RESEARCH

Reddit Data in Quantitative Financial Models (published) — Jan 2025 – May 2025
Co-author, Virginia Tech (VTechWorks) — *hdl.handle.net/10919/124730*

- Applied NLP sentiment scoring and regression models in Python to quantify sentiment's impact on equity returns.

PROJECTS

Casen | getcasen.com | *JavaScript, OpenAI, Supabase, Voice AI* — 2026

- Built a case-interview prep platform with frameworks, drills, M&A news, and voice-based AI mock interviews.

AI Quiz Maker | *Next.js, FastAPI, OpenAI* — 2026

- Built an app that turns PDFs and notes into quizzes, flashcards, and recall prompts using an LLM.

GreenSticker | greensticker.us | *Python, OpenCV*

- Built a hands-free webcam mouse that tracks a colored sticker, with low-latency on-device processing.

SPY Gradient Boosting Predictor | *scikit-learn, PCA, yfinance* — 2025

- Trained a gradient-boosting model to predict the next SPY close from EMA, RSI, and MACD features with PCA.

SKILLS

ML & AI: PyTorch, scikit-learn, XGBoost, YOLO11, OpenCV, LLM fine-tuning, LoRA, RAG, MCP, NLP, OpenAI API
Languages: Python, C++, C, SQL, Java, TypeScript, R, MATLAB
Data & Infra: NumPy, Pandas, SciPy, Linux, Docker, AWS, Azure, PostgreSQL, gRPC, FastAPI, Next.js, Power BI